Event Registration and Attendance Management Platform with Role-Based Access and Analytics

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***Abstract* - This paper presents an enhanced Event Registration and Attendance Management Platform designed as a scalable relational database system incorporating role-based access control, venue scheduling, ticketing, payments, session management, and analytical feedback mechanisms. The system enforces integrity constraints, prevents venue conflicts, manages event capacity with integrated waitlisting logic, and generates revenue and performance reports. The proposed architecture demonstrates advanced normalization and real-world event workflow modeling.**

Keywords - component, formatting, style, styling, insert (*key words****)***

I. INTRODUCTION

Institutions conduct workshops, seminars, hackathons, and cultural events that require structured coordination. Basic registration systems lack support for access control, financial tracking, scheduling constraints, and performance analytics.

This project extends traditional event databases into a multi-dimensional system modeling authority, time, space, and financial transactions within a unified relational framework.

II. SYSTEM DESIGN

- ***A. Role and User Management***
- ***1) Role***
- *role_id (PK)*
- *role_name (Student, Organizer, Admin, Volunteer)*
- ***2) User***
- *user_id (PK)*
- *name*
- *email*
- *password*
- *role_id (FK)*

- ***B. Event and Logistics***
- ***3) Venue***
- *venue_id (PK)*
- *venue_name*
- *building*
- *capacity*
- ***4) Event_Category***
- *category_id (PK)*
- *category_name*
- ***5) Event***
- *event_id (PK)*
- *event_name*
- *description*
- *event_date*
- *venue_id (FK)*
- *category_id (FK)*
- *organizer_id (FK → User)*
- ***C. Session Management***
- ***6) Event_Session***
- *session_id (PK)*
- *event_id (FK)*
- *session_name*
- *start_time*
- *end_time*

- ***D. Registration and Capacity Control***
- ***7) Registration***
- *registration_id (PK)*
- *user_id (FK)*
- *event_id (FK)*
- *registration_date*
- *registration_status (Confirmed / Waitlisted / Cancelled)*
- *waitlist_position*

- ***E. Attendance Tracking***
- ***8) Attendance***

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- *attendance_id (PK)*
- *registration_id (FK)*
- *session_id (FK)*
- *attendance_status*

- ***F. Ticketing and Payments***
- ***9) Ticket_Type***
- *ticket_id (PK)*
- *event_id (FK)*
- *ticket_name*
- *price*
- *quantity_available*
- ***10) Payment***
- *payment_id (PK)*
- *registration_id (FK)*
- *amount*
- *payment_status*
- *payment_date*
- *payment_method*

- ***G. Feedback and Analytics***
- ***11) Feedback***
- *feedback_id (PK)*
- *registration_id (FK)*
- *rating (1–5)*
- *comments*

III. Integrity Constraints

The system enforces:

1. Unique user-event registration
2. Venue conflict prevention
3. Capacity validation
4. Payment-registration consistency
5. Valid session attendance
6. Referential integrity across all tables

IV. Analytical Features

A. ***IV. ANALYTICAL FEATURES***

B. *Revenue per event*

C. *Most popular category*

D. *Attendance percentage per session*

E. *Average rating per event*

F. *Organizer performance metrics*

G. *Authors and Affiliations*

a) abbreviation "Fig. 1", even at the beginning of a sentence.

V. Conclusion

The refined schema balances complexity and manageability while demonstrating advanced relational modeling, constraint enforcement, and real-world business logic. The platform transitions from a basic registration database to a structured event management system capable of handling logistics, financial tracking, scheduling constraints, and analytical reporting.

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